

Vaibhav Singh

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Education

GL Bajaj Institute of Technology and Management, Greater Noida

2024 – 2028

B.Tech – Information Technology - CGPA 8.7/10

Coursework: Data Structures and Algorithms, Computer Organization and Design, Operating System, Calculus and Linear Algebra, Relational Database Management System, Artificial Intelligence & Machine Learning

Projects

Bhartiya Kanoon

React.js, Solidity, Polygon, MongoDB

- **Team Leadership:** Led a 4-member team to build a legal DApp ensuring secure document verification with strong data integrity (~99% immutability).
- **ML-Driven Classification:** Created classification/summarization pipeline using NLTK/SpaCy; improved document retrieval latency by ~8%.
- **Integrated Stack:** Connected MERN frontend with smart contracts (Polygon) and TensorFlow backend for real-time NLP. Education. [\[Live Demo\]](#)

TrustNet — Federated Learning over P2P Network

React.js, Node.js, Python, Django, Blockchain, Web3

- **Scalable FL Architecture:** Built a P2P federated learning system using a 6-ring poly-ring topology, enabling efficient routing and scalable participation across 20+ nodes.
- **Trust & Blockchain Layer:** Implemented trust-based node selection and smart-contract rewards, with a Django backend and UI for model selection, training orchestration, and blockchain-verified audit trails.[\[Live Demo\]](#)

Skills

- **Languages:** : C++, Python, JavaScript, TypeScript, PHP, Swift, HTML/CSS
- **Frameworks/Tools:** : React.js, Next.js, Node.js, Express.js, NestJS, FastAPI, LangChain, RAG, ElasticSearch, MCP, SwiftUI, Redux, MUI, Tailwind CSS, Git, Postman, VSCode, Figma
- **Databases/Security:** : PostgreSQL, MySQL, MongoDB, SQL, OWASP ZAP